Samson Wu

samson.wu@mail.utoronto.ca • linkedin.com/in/samson-wu-189979221/ • Dmoj
(647)-974-2570 • samson.systemprof.com • Toronto, Ontario

Summary

- **Objective**: Developer, Data Engineer, Data Modeler, Robotics Engineer, 3D designer, 2D Art – (Summer 2023 Intern)
- Passionate in programming, mechanical, and graphical design.

Skills

- Programming: Python, Pyspark, Java, SQL, PostgreSQL, Redshift database, C, HTML
- Software: AWS Cloud, AWS Glue, DB Visualizer, PyCharm, Git, Bitbucket, DevOps, Jenkins, Jira, Sharepoint, Agile.
- CAD/Simulations: Fusion360, Sketchup, MATLAB

Experience

Software Developer Co-op at Sunlife
May 2022 - August 2022 | Toronto, ON

- Assist with database upgrades, testing of Glue jobs, debugging, and code deployment using DevOps pipeline
- Participate in IFRS17 project, make Pyspark code change and testing to complete CMR requirements and defect fix
- Performed data modeling, data analysis on PostgreSQL and Redshift Database
- Completed the automation for partition table performance statistics using Pyspark, AWS S3, and Glue job
- Drafted GUI, created and populated Amazon RDS, and coded GUI in Python for Skills Matrix POC in a team.

Robotics Teams

- Captain of 2 FLL Teams (Falconators and Metal Talons) in 2013-2015. Research and Most Effective Design awards
- Captain of the Silver Stream Robotics Team achieving 2nd in the York Region Competition in 2016
- Led my team in the "Design, Build, Battle" UofT DEEP Program

Competitive programming

- Developed problem solving skills with 4 years of competitive programming using Java and Eclipse.
- Earned CCC Junior and Senior certificate of distinction (*top 25%*)

Structural Engineering
First year UofT Engineering Science

- Led the design and building of a matboard bridge for a structural engineering course. Achieved 4th strongest bridge in first year structural engineering course (out of 40+ teams)

Extracurriculars

Tennis for 10 years+
Competed in tournaments for high school team.

Thundercats Volleyball
Played in the team for 4 years as power and libero. Got 1st in 2018 Ontario Volleyball Tournament for under 15.

Bayview SS Volleyball
Captain of the grades 7 and 8 team. Communicated effectively and kept the team focused. Unsung Hero Award and 2nd Place in Regional Tournament.

Table Tennis
Played and took lessons for 6 years. Multiple podium placements.

Art
Took lessons for 5 years and have been drawing for 10+ years. 2nd in 2015 art competition *(300+ participants)*

Rubik's cubing for 9+ years
Top 141 in Canada for 3x3 Rubik's Cube time *(8.95 sec)*

Education

Engineering Science + PEY Co-op
University of Toronto
Sep. 2021 – 2026 (expected)

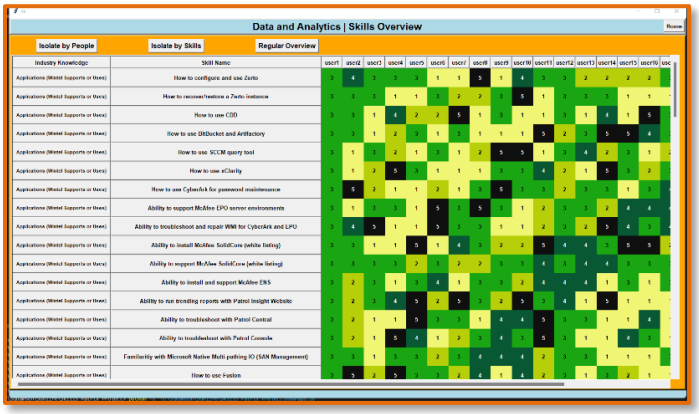
Bayview Secondary School IB Program
Sep. 2017 – *June 2021*

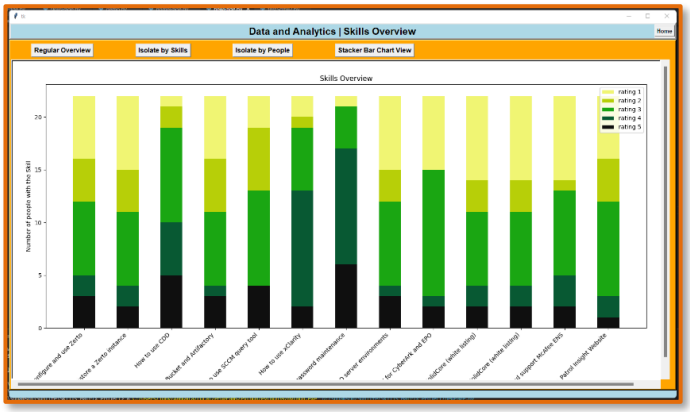
- 98.7% average in grade 12

Achievements

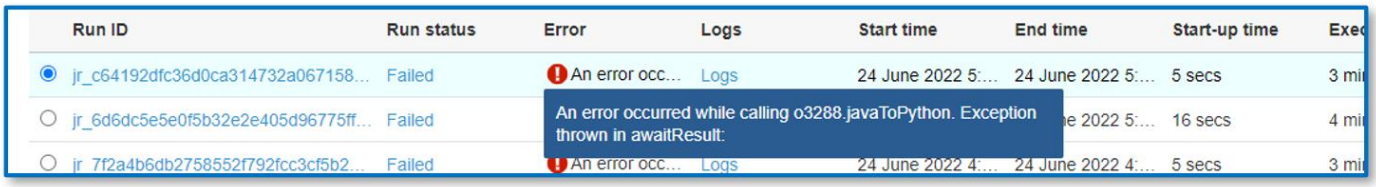
- Passed grade 9 piano
- First Class Honors with Distinction in Advanced Rudiments and Harmony

Portfolio (Software Development)



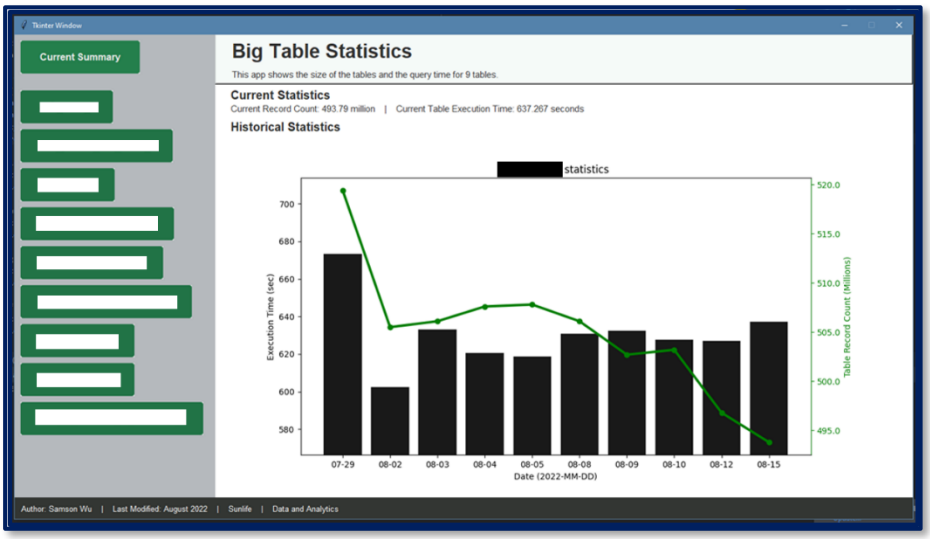


GUI for a Skills Matrix Project at Sunlife. Developed in Python as a proof of concept, with connection to an Amazon RDS database. The purpose of the app is to allow managers and employees to find skill gaps, monitor skill distribution and trends, and create new teams for new projects.



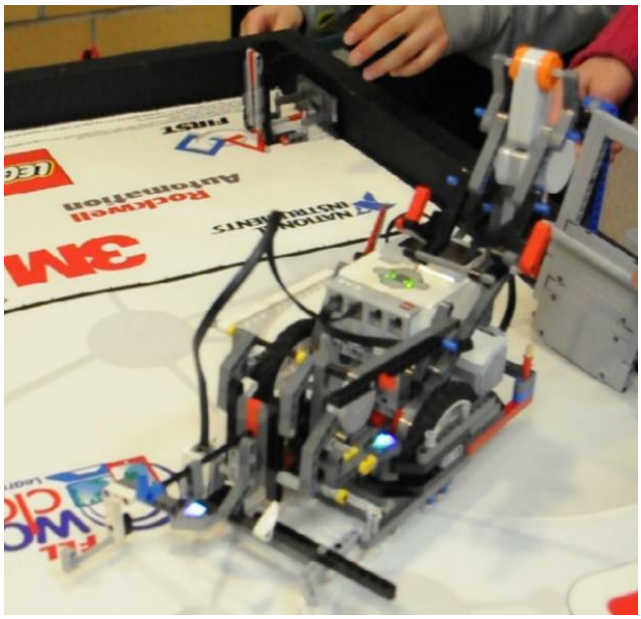

```
SELECT [redacted].file_name,
       [redacted].create_user_name,
       [redacted].batch_run_key,
       (MAX([redacted].step_date_and_time) - MIN([redacted].step_date_and_time)) as duration
FROM [redacted]
RIGHT JOIN [redacted]
ON [redacted].batch_run_key = [redacted].batch_run_key
WHERE [redacted].file_name in ('A02A0CLS_20220611030423.txt',
                               'A02AMFHC_20220624180044.txt',
                               'A02AMFHC_20220624180050.txt',
                               'A02AMFHC_20220624180051.txt',
                               'A02AMFHC_20220624180052.txt',
                               )
      AND [redacted].batch_run_key >= 8500
      AND [redacted].create_user_name LIKE '%enrichment'
GROUP BY [redacted].batch_run_key, [redacted].create_user_name
ORDER BY [redacted].batch_run_key ASC;
```

Analyzing AWS Glue Job performance and errors using PostgreSQL.



The GUI for my project at Sunlife. The objective is to display the details of large tables in the database for monitoring and analysis. The graph shows execution time for a query as black bars and row count as the green line. The interface and backend were developed in Python.

Portfolio (Robotics and Design)





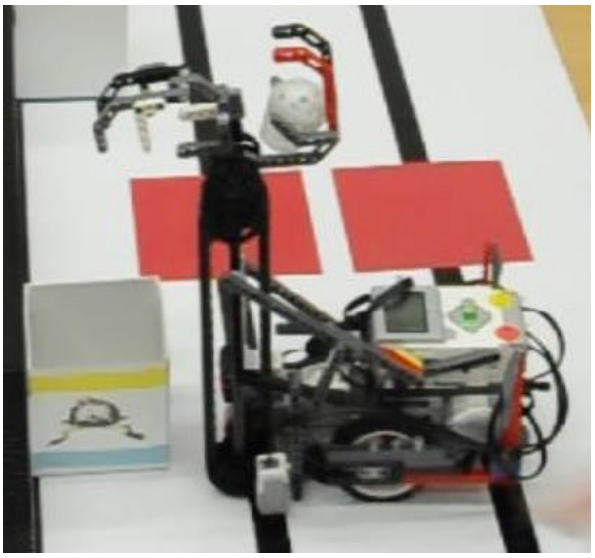
(Top left) FLL Robot for the 2014 challenge made to complete several missions or tasks.

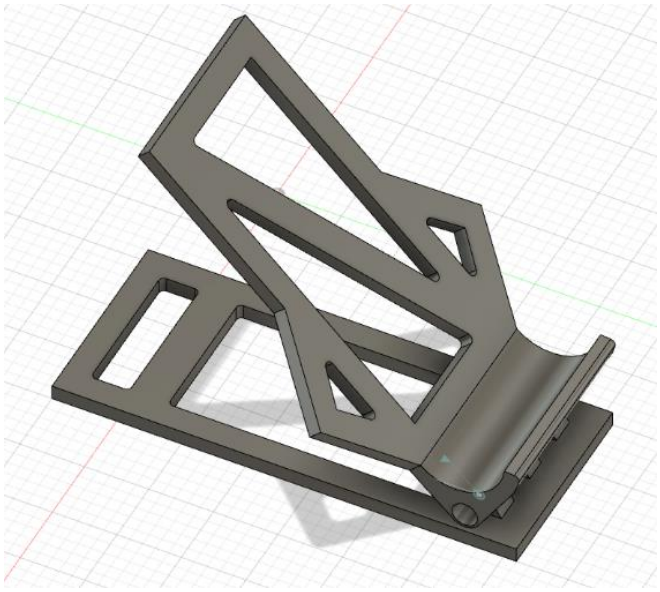
(Top right) structural engineering bridge design challenge.

(Right) York Region Robotics Robot made to put balls into boxes as well as pick up boxes.

(Bottom) Phone Stand design made with Fusion360.

(Bottom right) Richmond Hill Community Centre Design made with Sketchup.







Portfolio (painting and digital drawing)











